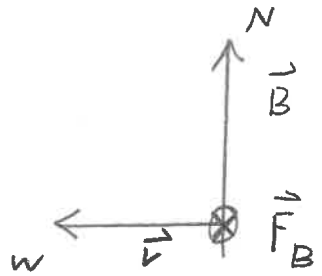
Chpt 18

(35.)



$B = 0.950\ T$

a) $q = 1.40\ \mu C$, $v = 1.50\ m/s$ (west)

$\vec{F}_B$ into page since $q > 0$

$F_B = |q| v B \sin\theta$, $\theta = 90^\circ$

$= qvB$

$= \boxed{1.995\ \mu N}$ (downward)

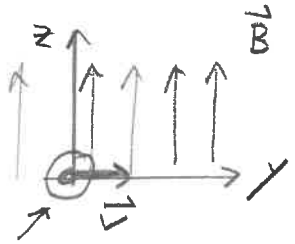
b) $q = -1.40\ \mu C$

$\vec{F}_B$ out of page since $q < 0$

$F_B = \boxed{1.995\ \mu N}$ (upward)

(43.) Proton $v = 4.50 \times 10^6\ m/s$ (y-direction)

$B = 450\ mT$



$\vec{F}_B$ (out of page; + x direction)

$\vec{F}_E$ must point into page (− x direction)

For zero net force, need $|q|E = |q| v B$

$E = vB = \boxed{2025\ \frac{N}{C}}$

(45.) Muon, $q = -e = -1.602 \times 10^{-19}\ C$

$v = 1.5 \times 10^6\ m/s \perp$ to $B = 1.2\ T$

Uniform circular motion with $R = 1.46\ mm$

$R = \frac{mv}{|q|B} \rightarrow m_\mu = \frac{|q| B R}{v} = \frac{1.602\times10^{-19} \times 1.2 \times 1.46\times10^{-3}}{1.5\times10^6}$

$= \boxed{1.87 \times 10^{-28}\ kg}$

$\rightarrow \frac{m_\mu}{m_e} = \boxed{205}$, $\frac{m_\mu}{m_p} = \boxed{0.112}$ (compared to mass of electron, proton)

(49.) proton: $T = 95\ \mu s$ For 1 complete orb. T

$D = 14.5\ m$ (diameter cyclotron)

recall: $T = \dfrac{2\pi m_p}{|q|B}$ (independent of R, v)

m_p = proton mass: 1.67×10^{-27} kg

$\rightarrow B = \dfrac{2\pi m_p}{|q|T} = \boxed{0.69\ mT}$

$|q|$ = proton charge: 1.602×10^{-19} C

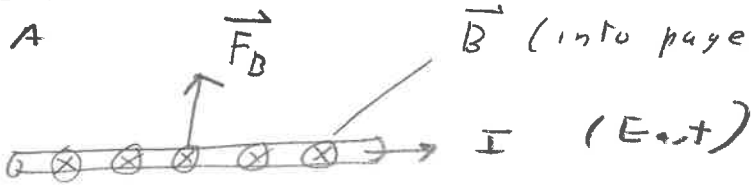
(59.) copper wire



$d = 0.150$ mm (diameter)

$L = 10.0$ cm

$I = 2.15$ A



$\vec{F}_B = I L B$ (upward)

$F_g = mg$ (downward)

$= \rho\ \pi r^2 \cdot L g$ where $\rho = 8.96\ g/cm^3 \left(\frac{100cm}{m}\right)^3 \left(\frac{1kg}{1000g}\right)$

$= 8960\ kg/m^3$

Thus, $I \not{L} B = \rho\ \pi r^2 \not{L} g$

$B = \dfrac{\pi r^2 \rho g}{I} = \boxed{722\ \mu T}$

(71.) $N = 1500$ turn

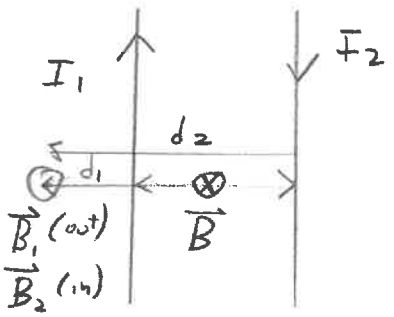
$d = 15$ cm diameter $\rightarrow r = 7.5$ cm (radius), $A = \pi r^2$

$\tau = \mu B$, $\mu = NIA = NI\pi r^2$

values: $\tau = 25\ N \cdot m$, $I = 12\ A$

$B = \dfrac{\tau}{\mu} = \dfrac{\tau}{NI\pi r^2} = \boxed{76\ mT}$

(75.)



$I_1 = I_2 = I = 2.5\,A$

$d = 1.5\ cm$

a) Halfway in between

$r = \frac{d}{2} = 0.75\ cm$

$\vec{B} = \vec{B}_1 + \vec{B}_2$ (both into page)

$\rightarrow B = \frac{\mu_0 I}{2\pi r} = \boxed{133\,\mu T}$

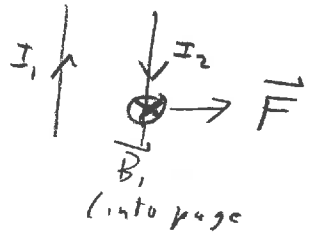
b) $B_1 = \frac{\mu_0 I_1}{2\pi d_1}$, $I_1 = I = 2.5A$, $d_1 = 1.5\ cm$

$B_2 = \frac{\mu_0 I_2}{2\pi d_2}$, $I_2 = I = 2.5A$, $d_2 = 3.0\ cm$

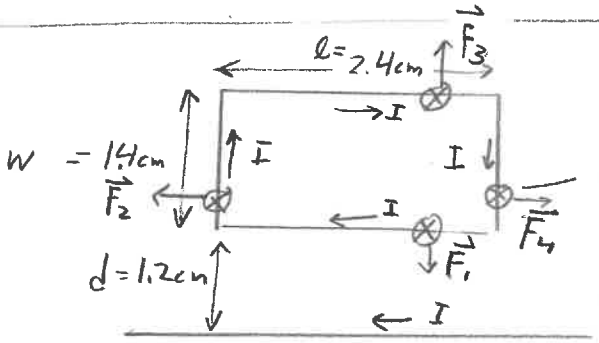
$B = B_1 - B_2 = \boxed{1.66 \times 10^{-5}\ T}$

c) $\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d} = \boxed{83\,\frac{\mu N}{m}}$

(repulsive)



(83.)



$I = 2A$, $d = 1.2cm$, $w = 1.4cm$, $\ell = 2.4cm$

magnetic field $\vec{B}$ from bottom wire is into page

$\vec{F}_{tot\ on\ loop} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4$

$= \vec{F}_1 + \vec{F}_3$ (since $\vec{F}_2 = -\vec{F}_4$ cancel)

$F_1 = I\ell B_1 = I\ell \frac{\mu_0 I}{2\pi d} = \mu_0 \ell \frac{I^2}{2\pi d}$

$F_3 = I\ell B_2 = I\ell \frac{\mu_0 I}{2\pi (d+w)} = \mu_0 \ell \frac{I^2}{2\pi (d+w)}$

a) $F_{on\ loop} = F_1 - F_3$ (toward wire)

$= \frac{\mu_0 \ell I^2}{2\pi}\left(\frac{1}{d} - \frac{1}{d+w}\right)$

$= \boxed{8.6 \times 10^{-7}\ N}$ (toward wire) } attractive

b) $F_{on\ wire} = \boxed{8.6 \times 10^{-7}\ N}$ (toward loop) } Use Newton's 3rd Law $\vec{F}_{AB} = -\vec{F}_{BA}$

c) If direction of current in rectangular loop is changed, then direction of Forces change

$F_{on\ loop} = \boxed{8.6 \times 10^{-7}\ N}$ (away From wire) } repulsive

d) $F_{on\ wire} = \boxed{8.6 \times 10^{-7}\ N}$ (away From loop)